

Exploring Health Insurance Costs

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1. Introduction

I used the **Insurance dataset**. It has 1,338 people. Each person has: age, sex, BMI, number of children, smoker or not, region, and insurance charges

I wanted to find out:

1. What makes insurance charges high?
2. Which features affect charges the most?
3. Any interesting or surprising patterns?

I also wanted to see how age, BMI, smoking, and number of children affect the costs, and if some people pay much more than others.

2. Cleaning Summary

The dataset had some problems:

- Missing values: I checked and there were no missing values, so all good.
- Data types: I changed sex, smoker, and region into categories so the model can understand them.
- Duplicates: I removed some duplicate rows to avoid counting the same person twice.
- Outliers: Some charges were very high. I capped them at the 99th percentile so they don't make graphs and analysis misleading.
- Check: I made sure all ages are positive, charges are not empty, and all columns are correct.

After cleaning, the data is ready to explore and use for models.

3. Feature Engineering Summary

I added some new columns to help understand the data better:

- BMI & children (BMI & child) – BMI divided by number of children. Helps see if weight per child matters.
- Obesity flag (fat) – 1 if BMI ≥ 30 , 0 otherwise. Shows who is obese.
- Age $\times$ BMI (elderly) – Multiply age and BMI. Older and heavier people usually pay more.
- Log of charges (ChargesLog) – To make very high charges less extreme and make distribution easier to see.
- Age group (age_group) – Child, young adult, adult, senior. Easier to see trends.
- One-hot encoding – Turn sex and region into numbers so models can use them.
- Scaling – age and BMI were scaled so numbers are in the same range.
- Remove very correlated columns – To avoid repeated information that confuses models.

4. Key Findings

Finding 1 — Smoking costs more

The boxplot of charges vs smoker shows smokers pay much more than non-smokers. Smokers also have more variation in charges. Smoking is a big factor in high costs.

Finding 2 — Age and BMI matter

Older people and people with higher BMI usually have higher charges. The elderly column (age × BMI) shows the combined effect. People with high BMI and older age pay the most.

Finding 3 — Most pay normal, few pay very high

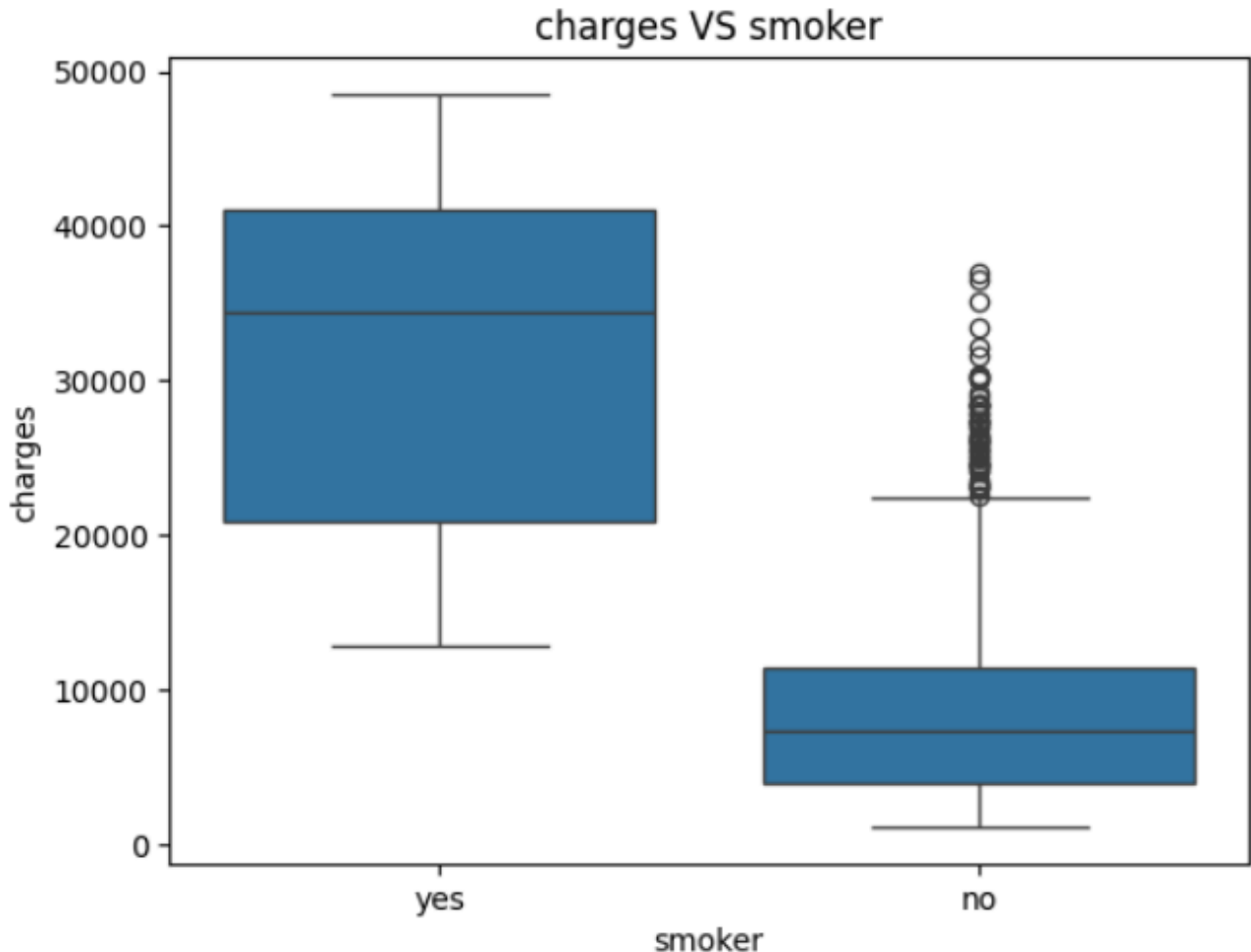
Histogram of charges shows most people pay medium costs. Only a small number pay very high charges. After log transform, the distribution looks more balanced, which is good for models.

Other patterns:

- Most people are adults (20–50), fewer are children or seniors.
- BMI is mostly normal (18–30), few very high or very low values.
- Region does not have big differences in charges.

Best chart:

- Charges vs Smoker boxplot – shows clear difference between smokers and non-smokers.



5. What I Would Do Next

If I had more time:

- Train a simple model with new features to predict charges.
- See if smoking affects older people more.
- Check regions with BMI and age together.
- Try more transformations for BMI and charges to help the model.
- Explore interactions like smoker $\times$ age or BMI $\times$ children.

Conclusion:

Smoking, age, BMI, and their combination are the main reasons charges go up. New features, scaling, and cleaning make the data ready for modeling and finding patterns. Most people have medium charges, but a few pay very high costs, which is important for insurance analysis.